

EE2213 Introduction to AI

Lecture 13: Propositional Logic

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AGENDA

- We will discuss:
 - Syntax of Propositional Logic
 - Semantics of Propositional Logic
 - Inference for Propositional Logic

At the end of this lecture, you should be able to:

- ✓ Explain the syntax and semantics of propositional logic
- ✓ Apply discussed inference methods to determine logical entailment in propositional logic

Propositional Logic: Introduction



➤ What is Propositional Logic?

A branch of logic where:

- Statements (propositions) are either **true** or **false**, but **not both**
e.g., $P = \text{"It is raining"}$, $Q = \text{"I have an umbrella"}$
- These statements can be combined using **logical connectives** (like AND, OR, NOT, etc.)
e.g., $P \text{ AND } Q = \text{"It is raining AND I have an umbrella"}$
- The truth value of complex statements can be determined from the truth values of their components

Propositional Logic: Introduction



➤ Importance of Propositional Logic

- ✓ Foundation for Formal Reasoning

A **formal, mathematical system** to express knowledge in a clear, unambiguous way.

- ✓ Building Block for Complex Logics

Basis for more expressive logical systems, such as first-order logic, description logic

- ✓ Enables Automated Inference

- ✓ Simplicity and Efficiency

Easy to implement and computationally tractable for small to medium-sized domains.

Propositional Logic: Syntax

➤ Proposition

A **statement/sentence** that is either true or false, but not both

Atomic proposition

A basic and indivisible statement

e.g., *“It is raining”*

(represented by a single proposition symbol, typically an uppercase letter like A, B, C, etc.)

Compound proposition

Built from atomic ones using **logical operators**.

e.g., *“It is raining AND I am indoors”*

Propositional Logic: Syntax

➤ Logical Connectives

Logical symbols that connect proposition symbols in order to reason in a more complex way about the world

- ✓ $\neg$ (**not**): A sentence such as $\neg P$ is called negation of P
e.g., P: “It is raining”, $\neg P$: “It is NOT raining”
- ✓ $\wedge$ (**and**): A sentence whose main connective is $\wedge$, such as $P \wedge Q$ is called a conjunction
e.g., P: “It is raining”, Q: “I bring an umbrella”.
 $P \wedge Q$: “It is raining AND I bring an umbrella”.
- ✓ $\vee$ (**or**): A sentence whose main connective is $\vee$, such as $P \vee Q$ is called a disjunction

➤ Logical Connectives

Logical symbols that connect proposition symbols in order to reason in a more complex way about the world

- ✓ $\Rightarrow$ (**implies**): A sentence such as $P \Rightarrow Q$ is called an **implication/conditional**. P is called **premise/antecedent**; Q is called the **consequent/conclusion**

e.g., P : “It is raining”, Q : “I am in doors”.

$P \Rightarrow Q$: “**IF** it is raining, **THEN** I am indoors”

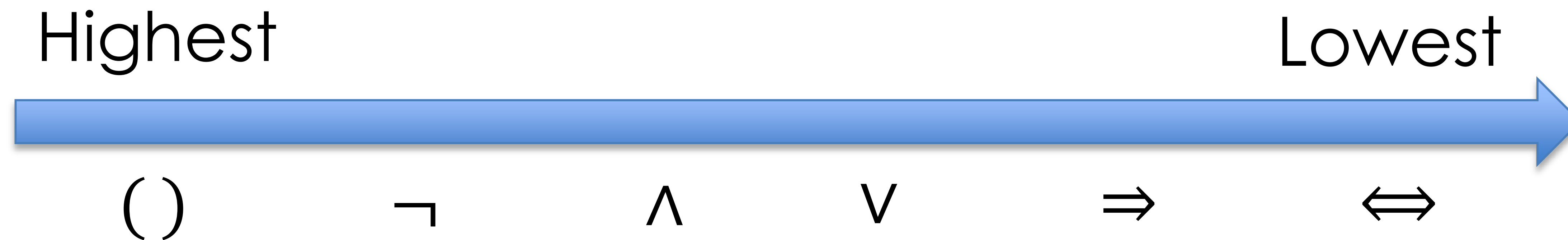
- ✓ $\Leftrightarrow$ (**if and only if**): A sentence such as $P \Leftrightarrow Q$ is called a **biconditional**

e.g., P : “It is raining”, Q : “I am in doors”.

$P \Leftrightarrow Q$: “**IF** it is raining, **THEN** I am indoors. **AND IF** I am indoors, **THEN** it is raining”

Propositional Logic: Syntax

➤ Operator Precedence



e.g., $\neg A \Leftrightarrow B \vee C \wedge D \Rightarrow E$

Propositional Logic: Syntax

➤ Well-formed Formula (WFF)

A valid expression built according to the syntactic rules:

1. Any proposition symbol is a WFF

2. If α is a WFF, then $\neg \alpha$ is a WFF

3. If α and β are WFFs, then the following are also WFFs:

$$(\alpha \wedge \beta), (\alpha \vee \beta), (\alpha \Rightarrow \beta), (\alpha \Leftrightarrow \beta)$$

e.g., Q1: If A and B are proposition symbols, $\neg(A \vee \neg B)$? ✓

Q2: $\neg(\alpha \vee \neg \beta)$? ✓

Q3: If A and B are proposition symbols, $A \wedge \vee B$? ✗

Propositional Logic: Semantics

➤ Rules for Determining the Truth

- Rules for computing the truth of **atomic propositions**

The truth value of every proposition symbol must be specified directly in the **MODEL**

↓
An assignment of truth value to **every** proposition symbol

e.g., P: “It is raining”, Q: “I am indoors”, R: “I bring an umbrella”

Model $m = \{P = \text{True}, Q = \text{False}, R = \text{True}\}$



m **satisfies** P (or m is a model of P)

$M(P)$: the set of all models of P

What about $m = \{P = \text{True}\}$?

*Not a valid model
(Can't determine Q and R)*

How many possible models in total?

Propositional Logic: Semantics

➤ Rules for Determining the Truth

- Rules for computing the truth of **compound propositions**

Rule 1: $\neg P$ is true **if and only if** P is false in model m

P	$\neg P$
True	False
False	True

e.g., P : “It is raining”, $\neg P$: “It is not raining”

Propositional Logic: Semantics

➤ Rules for Determining the Truth

- Rules for computing the truth of **compound propositions**

Rule 2: $P \wedge Q$ is true **if and only if** both P and Q are true in m

P	Q	$P \wedge Q$
True	True	True
True	False	False
False	True	False
False	False	False

e.g., P : “It is raining”, Q : “I bring an umbrella”,
 $P \wedge Q$: “It is raining and I bring an umbrella”.

Propositional Logic: Semantics

➤ Rules for Determining the Truth

- Rules for computing the truth of **compound propositions**

Rule 3: $P \vee Q$ is true **if and only if** at least one of them is true in m

P	Q	$P \vee Q$
True	True	True
True	False	True
False	True	True
False	False	False

e.g., P: “It is raining”, Q: “I bring an umbrella”,
 $P \vee Q$: “It is raining or I bring an umbrella”.

➤ Rules for Determining the Truth

- Rules for computing the truth of **compound propositions**

Rule 4: $P \Rightarrow Q$ is true **unless** P is true and Q is false in m

If P is true and Q is false, **then** $P \Rightarrow Q$ is false
otherwise, $P \Rightarrow Q$ is true

P	Q	$P \Rightarrow Q$
True	True	True
True	False	False
False	True	True
False	False	True

e.g., P : “It is raining”, Q : “I am indoors”,
 $P \Rightarrow Q$: “If it is raining, then I am indoors”.

➤ Rules for Determining the Truth

- Rules for computing the truth of **compound propositions**

Rule 5: $P \Leftrightarrow Q$ is true **if and only if** P and Q are both true or both false in m

P	Q	$P \Leftrightarrow Q$
True	True	True
True	False	False
False	True	False
False	False	True

e.g., P : “It is raining”, Q : “I am indoors”,
 $P \Leftrightarrow Q$: “If it is raining, then I am indoors.
And if I am indoors, then it is raining”.

Propositional Logic: Semantics

➤ Truth Table

A table that shows the **truth value** of a WWF for **every possible combination** of truth values of its atomic propositions

P	Q	$\neg P$	$P \wedge Q$	$P \vee Q$	$P \Rightarrow Q$	$P \Leftrightarrow Q$
true	true	false	true	true	true	true
true	false	false	false	true	false	false
false	true	true	false	true	true	false
false	false	true	false	false	true	true

Propositional Logic: Semantics

➤ Entailment (Logical Consequence)

A WWF α follows logically from another WWF β
(i.e., β entails α)

$$\beta \models \alpha$$

$\beta \models \alpha$ if and only if $M(\beta) \subseteq M(\alpha)$

(i.e., if β is true, then α must be true)

$\Rightarrow \beta$ entails α

e.g., $P \models (P \vee Q)$? ✓

$Q \models (P \Rightarrow Q)$? ✓

$(P \wedge Q) \models P$? ✓

Propositional Logic: Semantics

➤ Tautology

A WWF that is **always true** in all possible models.

e.g., $\neg P \vee P$

➤ Contradiction

A WWF that is **always false** in all possible models.

e.g., $\neg P \wedge P$

➤ Satisfiability

A WWF that is true/satisfied by *at least one* model

➤ Logical Equivalence

Two WWFs α and β are logically equivalent if they are true in the same set of models.

$\alpha \equiv \beta$ if and only if $\alpha \models \beta$ and $\beta \models \alpha$

$$(\alpha \wedge \beta) \equiv (\beta \wedge \alpha) \quad \text{commutativity of } \wedge$$

$$(\alpha \vee \beta) \equiv (\beta \vee \alpha) \quad \text{commutativity of } \vee$$

$$((\alpha \wedge \beta) \wedge \gamma) \equiv (\alpha \wedge (\beta \wedge \gamma)) \quad \text{associativity of } \wedge$$

$$((\alpha \vee \beta) \vee \gamma) \equiv (\alpha \vee (\beta \vee \gamma)) \quad \text{associativity of } \vee$$

$$\neg(\neg\alpha) \equiv \alpha \quad \text{double-negation elimination}$$

$$(\alpha \Rightarrow \beta) \equiv (\neg\beta \Rightarrow \neg\alpha) \quad \text{contraposition}$$

$$(\alpha \Rightarrow \beta) \equiv (\neg\alpha \vee \beta) \quad \text{implication elimination}$$

$$(\alpha \Leftrightarrow \beta) \equiv ((\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha)) \quad \text{biconditional elimination}$$

$$\neg(\alpha \wedge \beta) \equiv (\neg\alpha \vee \neg\beta) \quad \text{De Morgan}$$

$$\neg(\alpha \vee \beta) \equiv (\neg\alpha \wedge \neg\beta) \quad \text{De Morgan}$$

$$(\alpha \wedge (\beta \vee \gamma)) \equiv ((\alpha \wedge \beta) \vee (\alpha \wedge \gamma)) \quad \text{distributivity of } \wedge \text{ over } \vee$$

$$(\alpha \vee (\beta \wedge \gamma)) \equiv ((\alpha \vee \beta) \wedge (\alpha \vee \gamma)) \quad \text{distributivity of } \vee \text{ over } \wedge$$

Propositional Logic: Semantics

➤ Knowledge Base (KB)

A set of **WWFs** that represent **known** facts or rules about the world.

P: “It is raining”, Q: “I am indoors”, R: “I bring an umbrella”

$KB = \{P, Q, P \Leftrightarrow R\}$ P, Q, $P \Leftrightarrow R$ are assumed to be **true** in the KB.



$$KB = \{P \wedge Q \wedge (P \Leftrightarrow R)\}$$

What about $\{\neg P, P \wedge Q, P \Leftrightarrow R\}$?

Propositional Logic: Inference

➤ Model-checking Algorithm

1. Enumerate all the possible models *Check if $\alpha \models KB$*
2. Check if α is true in every model in which KB is true

e.g., P: "It is raining", Q: "I am indoors", R: "I bring an umbrella"

$KB = \{P, \neg Q, (P \wedge \neg Q) \Rightarrow R\}$

Query (Goal): $P \wedge R$?

P	Q	R	KB
false	false	false	
false	false	true	
false	true	false	
false	true	true	
true	false	false	
true	false	true	
true	true	false	
true	true	true	

Propositional Logic: Inference

➤ Inference Rule

Premise $\models$ Conclusion

Generate new information based on existing knowledge
without considering every possible model

Infer $\left(\frac{\alpha_1, \alpha_2, \dots, \alpha_k}{\beta} \right) \begin{matrix} \longrightarrow \text{Premise} \\ \longrightarrow \text{Conclusion} \end{matrix}$

e.g., α_1 : If it is raining, then I am indoors. α_2 : It is raining

What is β ?

I am indoors

1. Modus Ponens

$$\frac{\alpha \Rightarrow \beta, \alpha}{\beta}$$

2. And Elimination

$$\frac{\alpha \wedge \beta}{\alpha}$$

Propositional Logic: Inference

➤ Inference Rule

3. Logical Equivalence: $\alpha \equiv \beta$

Inference Rules:

$$\frac{\alpha}{\beta}, \frac{\beta}{\alpha}$$

$(\alpha \wedge \beta) \equiv (\beta \wedge \alpha)$	commutativity of $\wedge$
$(\alpha \vee \beta) \equiv (\beta \vee \alpha)$	commutativity of $\vee$
$((\alpha \wedge \beta) \wedge \gamma) \equiv (\alpha \wedge (\beta \wedge \gamma))$	associativity of $\wedge$
$((\alpha \vee \beta) \vee \gamma) \equiv (\alpha \vee (\beta \vee \gamma))$	associativity of $\vee$
$\neg(\neg\alpha) \equiv \alpha$	double-negation elimination
$(\alpha \Rightarrow \beta) \equiv (\neg\beta \Rightarrow \neg\alpha)$	contraposition
$(\alpha \Rightarrow \beta) \equiv (\neg\alpha \vee \beta)$	implication elimination
$(\alpha \Leftrightarrow \beta) \equiv ((\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha))$	biconditional elimination
$\neg(\alpha \wedge \beta) \equiv (\neg\alpha \vee \neg\beta)$	De Morgan
$\neg(\alpha \vee \beta) \equiv (\neg\alpha \wedge \neg\beta)$	De Morgan
$(\alpha \wedge (\beta \vee \gamma)) \equiv ((\alpha \wedge \beta) \vee (\alpha \wedge \gamma))$	distributivity of $\wedge$ over $\vee$
$(\alpha \vee (\beta \wedge \gamma)) \equiv ((\alpha \vee \beta) \wedge (\alpha \vee \gamma))$	distributivity of $\vee$ over $\wedge$

e.g.,

$$\frac{\alpha \Leftrightarrow \beta}{(\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha)}$$
$$\frac{(\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha)}{\alpha \Leftrightarrow \beta}$$

➤ Inference Rule

4. Resolution Rule

$$\frac{\alpha \vee \beta, \neg \alpha}{\beta}$$

e.g., $\alpha \vee \beta$: Sandy is in the canteen or library.

$\neg \alpha$: Sandy is not in the canteen.

$$\frac{\alpha_1 \vee \cdots \vee \alpha_n, \quad \gamma_1 \vee \cdots \vee \gamma_m}{\alpha_1 \vee \cdots \vee \alpha_{i-1} \vee \alpha_{i+1} \vee \cdots \vee \alpha_n \vee \gamma_1 \vee \cdots \vee \gamma_{j-1} \vee \gamma_{j+1} \vee \cdots \vee \gamma_m}$$

where α_i and γ_j are complementary **literals**

(Literal: atomic proposition or negated atomic proposition)

➤ Inference Rule

4. Resolution Rule

$$\alpha_1 \vee \cdots \vee \alpha_n, \quad \gamma_1 \vee \cdots \vee \gamma_m$$

$$\alpha_1 \vee \cdots \vee \alpha_{i-1} \vee \alpha_{i+1} \vee \cdots \vee \alpha_n \vee \gamma_1 \vee \cdots \vee \gamma_{j-1} \vee \gamma_{j+1} \vee \cdots \vee \gamma_m$$

Clause: a disjunction of literals

Can it be applied to other WWFs for premises (such as $\Rightarrow$, $\Leftrightarrow$)?

Every proposition $\equiv$ A conjunction of clauses

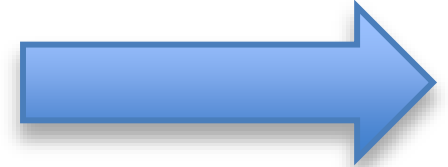
(Conjunctive Normal Form (CNF))

$$Clause_1 \wedge \cdots \wedge Clause_n$$

e.g., Convert $A \Leftrightarrow (B \vee C)$ into CNF

Propositional Logic: Inference

➤ Resolution Algorithm

Check if $KB \models \alpha$  Check if $KB \wedge \neg \alpha$ is contradiction

Step 1. Convert $KB \wedge \neg \alpha$ into CNF

Step 2. Apply resolution rule to each pair of clauses that contains **complementary literals** to produce a new clause, which is added to the KB if it is not already present, until one of the following happens:

- No new clauses can be produced anymore.

No entailment!

- *Empty* clause is produced (equivalent to False)

$KB \models \alpha$

Propositional Logic : Inference

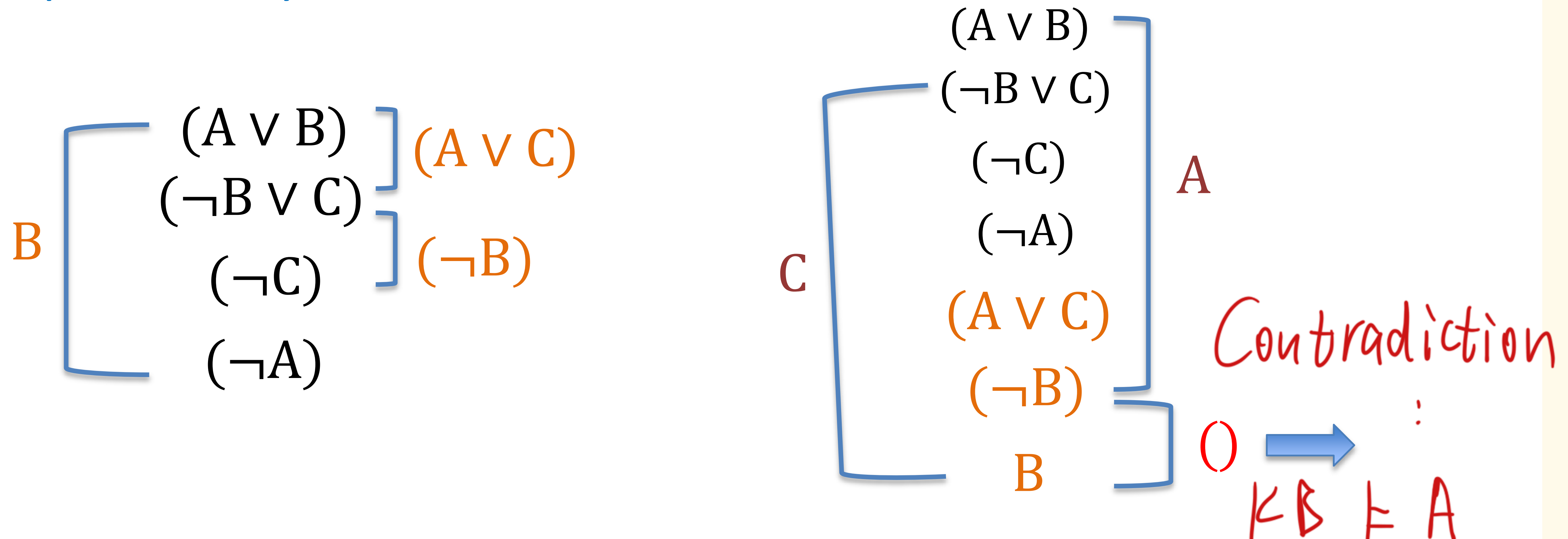
➤ Resolution Algorithm

e.g., does $KB = \{A \vee B, \neg B \vee C, \neg C\}$ entail A ?

Step 1. Convert $KB \wedge \neg A$ into CNF

$$KB \wedge \neg A = (A \vee B) \wedge (\neg B \vee C) \wedge (\neg C) \wedge (\neg A)$$

Step 2. Apply resolution rule to each pair of clauses that contains complementary literals



Propositional Logic : Inference

➤ Resolution Algorithm

Important!

Empty clause produced must arise only from resolving **two single literals (unit clauses)**

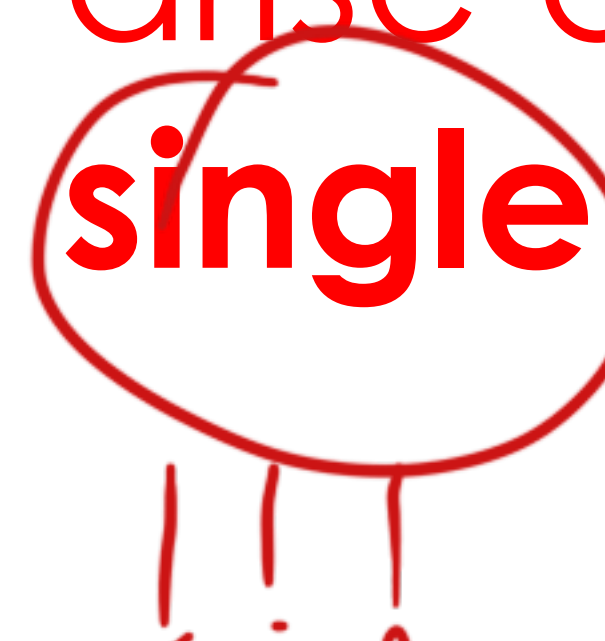
e.g.,

$A \vee B, \neg A \vee \neg B$

?



Multiple literals



$A, \neg A$

?

Propositional Logic : Inference

➤ Forward Chaining Algorithm

Infer atomic propositions based on KB of **definite clauses**.

A disjunction of literals of which exactly one is positive

e.g., $(\neg A \vee \neg B \vee C)$

Step 1: Start with atomic propositions in the KB

Step 2: Apply inference rules to infer new facts and add them to the KB

Step 3: Continue until:

- The goal (query) is derived, or
- No more new facts can be inferred

Propositional Logic : Inference

➤ Forward Chaining Algorithm

Step 1: Start with atomic propositions in the KB

Step 2: Apply inference rules to infer new facts and add them to the KB

Step 3: Continue until: -The goal (query) is derived, or
-No more new facts can be inferred

e.g., $KB = \{A, B, A \wedge B \Rightarrow C, A \wedge P \Rightarrow L, B \wedge C \Rightarrow P\}$, Query: L

Step 1: A is true. B is true

Step 2: From $A \wedge B \Rightarrow C$, infer C is true.

Update $KB = \{A, B, A \wedge B \Rightarrow C, A \wedge P \Rightarrow L, B \wedge C \Rightarrow P, C\}$

From $B \wedge C \Rightarrow P$, infer P is true.

Update $KB = \{A, B, A \wedge B \Rightarrow C, A \wedge P \Rightarrow L, B \wedge C \Rightarrow P, C, P\}$

From $A \wedge P \Rightarrow L$, infer L is true. ➡

Propositional Logic : Inference

➤ Backward Chaining Algorithm

Infer atomic propositions based on KB of **definite clauses**.

Start with the query (goal)

Step 1: Find implications in the KB whose conclusions are query.

Step 2: Try to prove the premises of those implications

Step 3: Recursively prove subgoals until reaching the atomic propositions in the KB

Propositional Logic : Inference

➤ Backward Chaining Algorithm

Step 1: Find implications in the KB whose conclusions are query.

Step 2: Try to prove the premises of those implications.

Step 3: Recursively prove subgoals until reaching the atomic propositions in the KB

e.g., $KB = \{A, B, A \wedge B \Rightarrow C, A \wedge P \Rightarrow L, B \wedge C \Rightarrow P\}$, Query: L

Step 1: Find $A \wedge P \Rightarrow L$ whose conclusion is L, check if $A \wedge P$ is true.

Step 2: A is true; Find $B \wedge C \Rightarrow P$ whose conclusion is P, check if $B \wedge C$ is true.

B is true; Find $A \wedge B \Rightarrow C$ whose conclusion is C

$A \wedge B$ is true 

Summary

✓ Syntax

Propositions, Logical Connectives, Well-formed Formula (WFF)

✓ Semantics

Truth Table, Model, Entailment, Tautology, Contradiction, Satisfiability, Logical Equivalence, Knowledge Base

✓ Inference

Model Checking Algorithm, Resolution Algorithm, Forward Chaining Algorithm, Backward Chaining Algorithm.